# EE 5290 / Data Analytics in Electrical and Computer Engineering

Iowa State University

## Contact info

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| Instructor | Aleksandar Dogandžić | Coover 3119 |
| | ald@iastate.edu | Office hours: M W 2–3 or by appointment |

## Prerequisites

Familiarity with linear algebra and coding/programming. EE 3220 or equivalent course in probability and random processes. Familiarity with optimization will be a plus.

## Description

This course offers an introduction to data analytics and covers topics relevant to electrical and computer engineers.

## Objectives

Upon successful completion, you will be able to:

- grasp the basics of modern data analytics and machine learning,
- mathematically model problems in data analytics arising in real-world applications,
- propose algorithmic solutions to these problems using a variety of tools,
- analyze these algorithms in terms of their efficiency and correctness, and
- implement and test software prototypes of several algorithms of interest.

## Textbook

We will mostly follow course lecture notes and slides.

Supplemental textbooks are "Probabilistic Machine Learning: An Introduction" by Murphy (containing Python examples) and "Foundations of Data Science" by Blum, Hopcroft, and Kannan (mathematically oriented). A good undergraduate-level book with relevant content on clustering and least-squares regression and classification is “Introduction to Applied Linear Algebra” by Boyd and Vandenberghe. Some topics are also covered in “The Elements of Statistical Learning,” by Hastie, Tibshirani, and Friedman and in Leskovec et al. These references are free.

Programming aspects (ISU free access, code on Github): Raschka & Mirjalili, *Python Machine Learning: Machine Learning and Deep Learning with Python, scikit-learn, and TensorFlow 2*; Müller & Guido, *Introduction to Machine Learning with Python*.

# Venue

Lectures: M W 4:25–5:45 in Coover 1012

## Topics (tentative)

- Math basics, inner product, vectors, matrices,
- Modeling data in high dimensions,
- Regression: linear, logistic,
- Classification: nearest neighbors, SVMs, kernel methods, neural nets,
- Visualization and Dimensionality Reduction: PCA, clustering, Isomap, MDS,
- Graphs: Random walks, PageRank,
- Big Data: Compressed sensing and matrix completion,
- Introduction to streaming algorithms.

## Grading policy

- 60 % Assignments
- 10 % Project proposal
- 10 % Project presentation
- 20 % Project report

## Homework

Homework-problem sets will be handed out during the semester. Problem sets will involve a mix of theoretical analysis and practical implementation. Theoretical problems may involve a short mathematical derivation, an analysis of a particular data processing technique, or a construction of a new method.

Implementation problems will involve some degree of programming. You are free to use a scientific programming environment that you are most comfortable with. Python/Matlab/R/Julia will suffice for most problems.

I *strongly* recommend using Python, even if you aren't familiar with it. There are several reasons for this:

- Python has become the programming language of choice for data analytics and machine learning – particularly for engineering-based applications. Supporting packages/libraries for Python are also fairly mature and stable.
- Python has a fairly gradual learning curve and is easy to pick up.
- Python pushes you to write neat and readable code, and makes it easier for everyone (us) to understand (grade).

Bottom line: Use Python if at all possible. Try to submit code using Jupyter Notebooks (Python, but also Julia & R) with self-contained explanations and results.

Any programming-assignment solution needs to show results of running the code, and discuss those results.

Any theoretical-assignment solution needs explanation. There could be exceptions for very small credit. If a (sub)question is worth 1 point, explanation may not be needed, but it is still suggested.

## Collaboration policy

You are encouraged to collaborate with your class mates on homework assignments. However, please (a) clearly acknowledge your collaborator/s, and (b) compose your final writeup (including **all** derivations and code) by yourself.

If two assignments are obviously identical to each other, then **both** will receive a score of zero. Please talk to me in advance for any clarifications.

No late submissions please! Your grade will be reduced by 25% each day it is late.

## Course project

Final exam consists of a course project. The goal of the project is open-ended; it can involve either: (a) conducting research on a specific topic of your choice, or (b) coding up a technique and testing it on a real-world dataset, or (c) both. The only requirement is that it should involve (any combination of) analysis, design, or implementation of a data analytics technique.

Projects are conducted in groups of at most two (2). You'll have to come up with project ideas *yourself*. It will be particularly beneficial for you if you can integrate the project with your own research interests, or align it with a problem or project that you want to learn more about. Start thinking of project ideas early and discuss them with me before finalizing.

For the project, there will be three deliverables:

- A two-page project proposal due on March 25.
- A project presentation, to be given on May 4 and 6, i.e., **during the Prep Week** and likely also the week before, April 27 and 29. Please plan accordingly.

- A project report of no more than 12 pages including all figures and references, to be turned in on May 14.

## Student accommodations

If you have a documented disability and anticipate needing accommodations in this course, please meet with me. Please request that a Disability Resources (DR) staff send a Student Academic Accommodation (SAAR) form verifying your disability and specifying the accommodations you need. DR is located in Room 1076 of Student Services.

Last edited: December 13, 2025